Conversion of Camphene to Camphor D3

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28 November 23

CHEM 2081-003

Introduction

On day three of the Conversion of Camphene to Camphor experiment, borneol was converted to camphor. Once the reaction was completed, the collected crystals were weighed and the mass was recorded.

Materials and Methods

A sand bath was plugged in and set aside to heat up. In a long neck round bottom flask 0.3 g of borneol product from day two was added with 0.3 mL of acetic acid. This racemic mixture was then dissolved in 1 mL of acetone. A reflux with a distillation column was assembled and the apparatus was placed in a water bath that was set on the sand bath. Once all of the reagents were dissolved, 6 mL of aqueous sodium hypochlorite solution was added dropwise over 35 minutes. The reaction continued for 15 minutes after the last addition. Next the mixture was left to cool and then the camphor was extracted with two portions of methylene chloride. The combined methylene chloride was then neutralized by 2 mL of saturated sodium bicarbonate. Then, the organic solution was washed with 2 mL of sodium bisulfate and 2 mL of water. The organic solution was dried over anhydrous sodium sulfate. Finally, the DCM was evaporated over a water bath and the collected crystals were weighed.

Results

Amount of Product (g)	0.31

Table 1. Amount of End Product

Calculations

Theoretical Yield:

$$(0.3\ g\ borneol) \times (\frac{1\ mol\ borneol}{154.25\ g\ borneol}) \times (\frac{1\ mol\ camphor}{1\ mol\ borneol}) \times (\frac{152.23\ g\ camphor}{1\ mol\ camphor}) = 0.297\ g\ camphor$$

Percent Yield:

$$\left(\frac{0.31\,g}{0.297g}\right) \times 100\% = 104.36\%$$

Discussion

In this experiment, about 0.31 g of end product was produced through converting borneol to camphor. Since solid was left over after evaporation was complete, the product is most likely camphor. One source of error for the percent yield being over 100% could be that the drying agent was not completely removed. This would lead to an overestimated yield of product and give inaccurate results. Another possible source of error that could have occurred was stopping the reflux too early. The most likely source of error to have occurred was not removing all of the drying agent before evaporation. This would lead to a higher yield since there was more end product than sample that was started with. Since 104.36% of the original sample was produced after the reaction, this was the most likely error to have occurred.

References

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry 2 Laboratory Manual [lab manual]. Auburn (AL): Auburn University.

Lab Performance

Exp 6 D3

Name: *Section:* *Room:* *Hood:*

Wearing personal protection equipment—this evaluation records your consistency with wearing the appropriate clothes for lab (long pants, closed toe shoes, and tops with sleeves) as well as the required laboratory safety equipment of apron, goggles, and gloves. ○ ○ ○ ○ ○ (5pts)

Safe working habits—as the GTAs do the lab rotations, they will record your work ethic and whether you follow our safety rules and regulations concerning tied hair, tied apron, safety goggles on at all times, and no food and drinks. ○ ○ ○ ○ ○ (5pts)

Following our laboratory safety technique work procedures—evaluation of operation of our laboratory equipment—water sinks in hood (1pt), vacuum and air lines (1pt), sash position (1pt), and heating elements (1pt). It is important that after every experiment you shut down the hood and make it ready for the section after yours (5pts). ○(1pt) ○(1pt) ○(1pt) ○ (1pt) ○ (5pts)

Glassware operation—safe handling of our glassware equipment. This means cleaning (1pt), maintaining the drawer inventory (5pts), and proper experimental setup (5pts). ○(1pt) ○ (5pts) ○ (5pts)

Working independently or equal partner participation during the entire lab. ○ ○ ○ ○ ○ (5pts)

Good work behavior—you work in a dynamic environment; therefore, creating distractions through noise and disobedience will not be tolerated. No cell phone use, listening to loud music, or working/distracting at a non-assigned workspace. ○ ○ ○ ○ ○ (5pts)

Maintain clean and organized personal space. ○ ○ ○ ○ ○ (5pts)

GTA notes:

GTA signature: _____ *Final Performance Grade:* [ ]

Exp. No. Le D3	Experiment/Subject camphene to Camphor	Date 11/14	57
Name Caroline McCoy	Lab Partner	Locker/ Desk No.	Course & Section No. 003

(1) LNRBF 10mL add 0.3g borneol product w/ 0.3mL glacial acetic acid
- Dissolve in 1mL acetone

- rxn performed in water
- aqueous bleach below
- DCM will sit ~~below~~ rxn mixture
- DCM D = 1.33 g/mL
- exothermic rxn

(2) Like D2 reflux
- If borneol doesn't dissolve add 0.5mL acetone to mixture until all borneol goes into soln

WG: 33.29
WG+P: 33.598

(3) once dissolved add 6mL of aqueous sodium hypochloride soln
- dropwise over 35min
- swirl
- 15 mins after last adn

just add slowly

(4) let mixture cool to RT & extract camphor w/ 2 portions (2mL each) of methylene chloride DCM

(5) neutralize methylene chloride layers w/ ~2mL saturated sodium bicarbonate

- if it keeps bubbling add more
- take pH w/ red LP

(6) wash organic soln w/ 2mL 5% sodium bisulfite & 2mL H_2O

- organic layer

(7) Dry organic soln over sodium sulfate dry when free rolling crystals

DCM = bottom layer

(8) Evaporate DCM over water bath w/ temp not exceeding 50°C. Weigh crystals
- percent yield
- melting point

- hot bath DCM on watch g

Signature	Date	Witness/TA	Date

KENDALL-McNEIL STUDENT LAB NOTEBOOK Note: Place fold-over back cover under copy sheet before